THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 5 & 6: **Exploring Forces, Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Mass is measured in: 1

(a) metres **(b)** newtons **(c)** kilograms **(d)** pascals

2. A body of mass 1 kg has weight ___ on the Moon. 1

(a) 25.4 N **(b)** 9 N **(c)** 10 N **(d)** 1.6 N

3. In a spring balance, 1 N is divided into 5 equal parts. Each division is: 1

(a) 0.2 N **(b)** 0.5 N **(c)** 2 N **(d)** 0.1 N

4. An object sinks when gravitational force is: 1

(a) less than buoyant force **(b)** absent

(c) equal to buoyant force **(d)** greater than buoyant force

5. Muscular force helps in: 1

(a) only walking **(b)** chewing food and pumping blood

(c) only jumping **(d)** only lifting

6. During the day, air over land rises because land: 1

(a) stays cold **(b)** is wet

(c) heats faster than water **(d)** cools faster than water

7. Lightning is caused by: 1

(a) dust **(b)** wind alone

(c) sunlight **(d)** sudden flow of built-up static charges

8. Safe action during lightning is to: 1

(a) stand under a tree

(b) use a metal umbrella

(c) lie flat on the ground

(d) crouch low in an open area away from tall objects

9. A cyclone weakens after reaching land because: 1

(a) ocean returns **(b)** land is cold

(c) moist air supply is cut off **(d)** wind turns stronger

10. Mango showers occur in: 1

(a) Assam only **(b)** Punjab

(c) Bihar only **(d)** Kerala, Karnataka and Tamil Nadu

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. A 3 kg mass has weight about how much on Earth? (Take 1 kg = 10 N). 2

12. State the use of a spring balance. What does its range mean? 2

13. Why do two balloons come close when we blow between them? 2

14. Why are roofs blown away during high-speed winds? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain why weight differs on different planets while mass does not. 3

16. Why does a cyclone weaken on reaching land? 3

17. Give three examples of muscular force in daily life and in the body. 3

OR

How is friction useful? Give three examples and say what would happen without it. 3

18. State three local thunderstorms of India and where they occur. 3

OR

Why are the straps of heavy bags made wide? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Compare the three non-contact forces in a table with nature and one example each. 5

OR

Explain why a heavy ship floats and a small iron nail sinks, using upthrust. 5

20. A tank 3 m high is on a roof. Explain why the top floor gets a weak stream. Also calculate pressure when a force of 900 N acts on 0.3 m². 5

OR

Compare a thunderstorm and a cyclone on formation, location, effects and duration. 5

END OF QUESTION PAPER — Total: 40 Marks

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